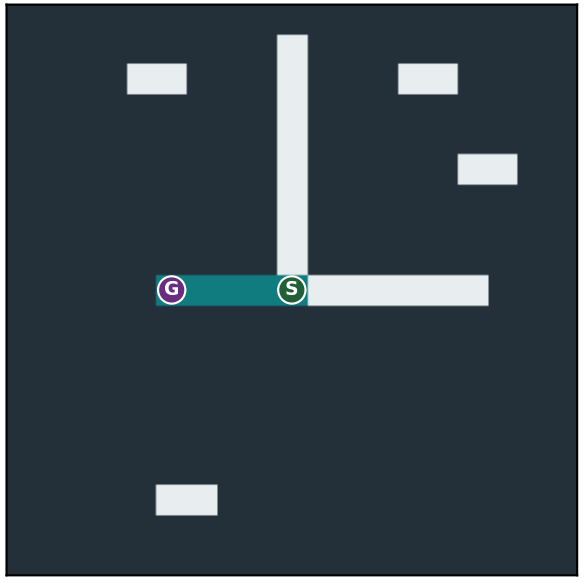
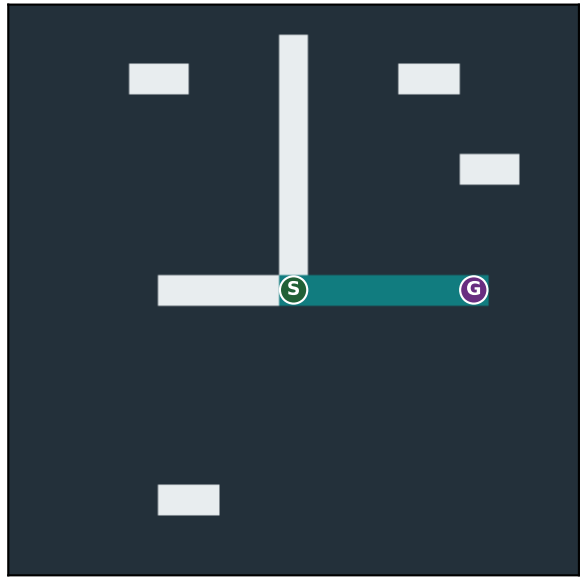
Same unseen fork: move the source one cell upstream
Actual emitted rasters — teal is correct, coral is unsuccessful. S = source, G = goal.

Mutated descendant

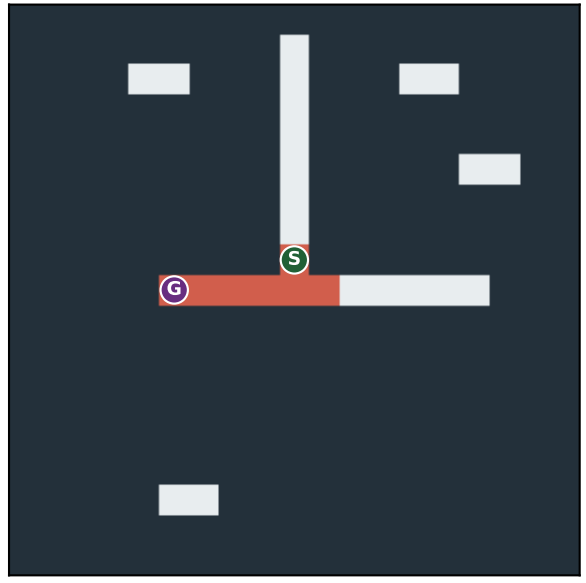
Prefix 0, goal A
Shortest path



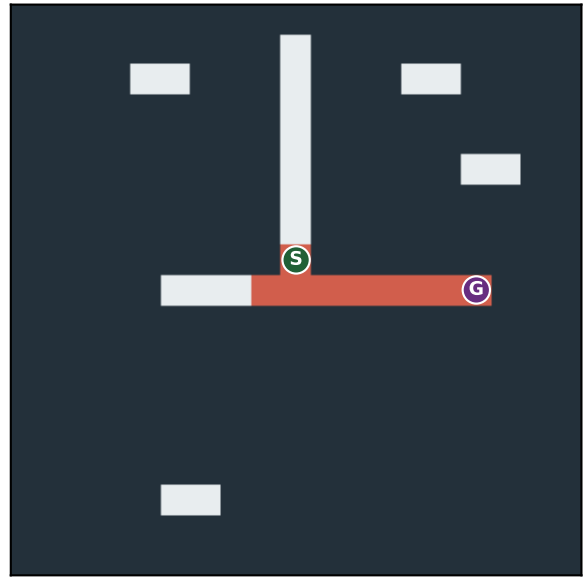
Prefix 0, goal B
Shortest path



Prefix 1, goal A
branch or cycle

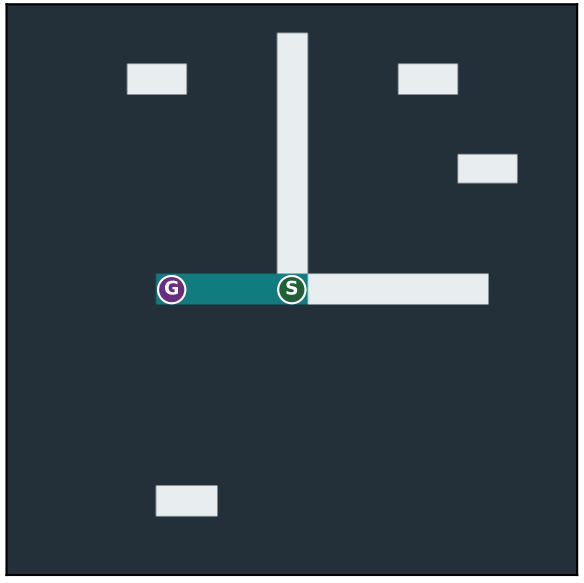


Prefix 1, goal B
branch or cycle

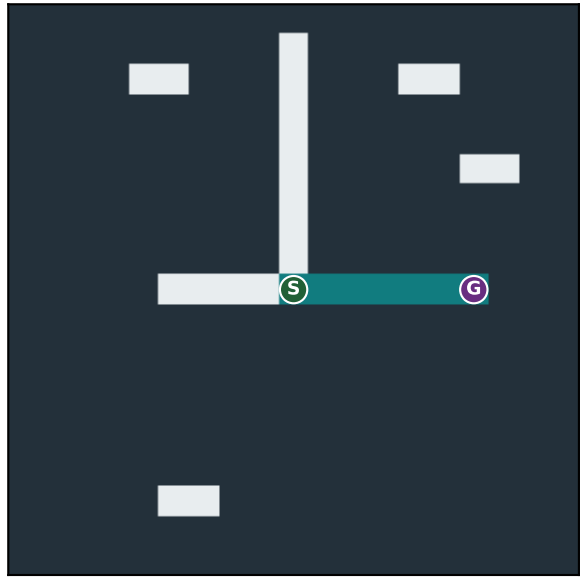


Unmutated random draw

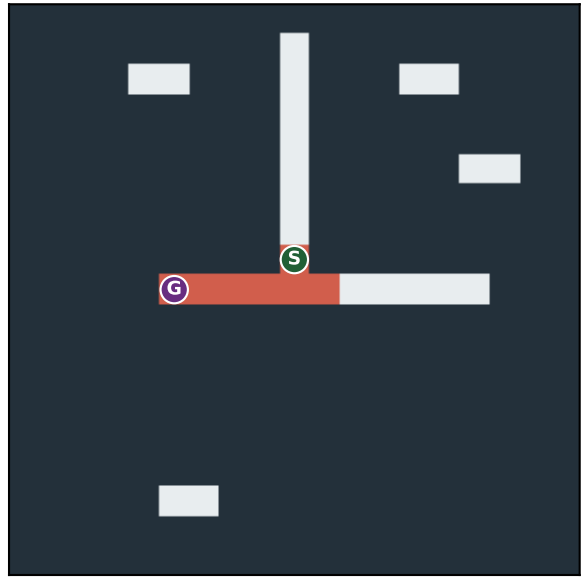
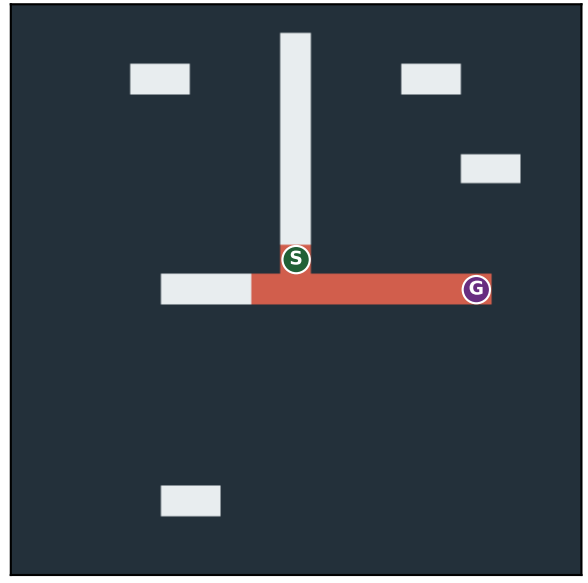
Prefix 0, goal A
Shortest path



Prefix 0, goal B
Shortest path



Prefix 1, goal A
branch or cycle



Prefix 1, goal B
branch or cycle